

Microservice architecture for a modern web application:

The system starts with a user accessing a web frontend built in React.

The frontend communicates with an API Gateway which routes all requests.

The API Gateway forwards authentication requests to an Auth Service.

The Auth Service validates users and issues tokens.

For user data operations, requests are sent to a User Service.

The User Service interacts with a dedicated User Database.

For file uploads, the API Gateway sends requests to an Upload Service.

The Upload Service stores files in an Object Storage system.

A Processing Service analyzes uploaded data and performs computation tasks.

The Processing Service communicates with a Message Queue for asynchronous tasks.

Finally, results are stored in a Results Database and returned to the frontend via the API Gateway.